

★ WEEK 15 · PART 3 of 3

Bayesian Switching: Regimes, Breaks, and Stage VI

Soft Regime Calls and Trading With Uncertainty

Bayesian HMM · structural breaks · label switching · the capstone

We re-estimate the regime model of Week 10 with priors, get soft credible regime probabilities instead of hard labels, locate structural breaks at uncertain dates, and complete the pairs-trading thread at Stage VI.

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PART 3 · BAYESIAN SWITCHING

Regimes, Breaks, and the Capstone

The Bayesian HMM and soft regime probabilities, structural breaks at uncertain dates, the label-switching gotcha, sizing positions by conviction, and pairs trading Stage VI.

From Drifting States to Switching States

Parts 1-2 drifted; Part 3 switches

1

We handled continuous drift

Parts 1-2: hedge ratios and volatilities that drift smoothly, with FFBS drawing whole paths and honest bands.

2

But markets also switch abruptly

Calm flips to crisis; a relationship breaks at a date. These are discrete regimes and jumps, not smooth drift.

3

Bayes handles switching too

The Bayesian HMM gives soft regime probabilities; break models locate jumps with uncertainty. Both feed position sizing.

Part 3 in one line: soft regime probabilities and uncertain break dates, flowing through to how much you trade.

The Bayesian HMM

Re-estimating Week 10's regime model with priors

Week 10 fit a hidden Markov model by Baum-Welch — a point estimate of the transition matrix and the regime parameters. The Bayesian HMM puts priors on all of it and infers a full posterior, so every regime quantity comes with honest uncertainty.

What Week 10 gave

- A single best transition matrix.
- Point estimates of each regime's parameters.
- Hard-ish regime assignments via the filter.

What Bayes adds

- Priors on the transitions and emissions.
- A posterior over transitions, parameters, and states.
- Credible bands on everything, regimes included.

The Bayesian HMM Posterior

Transitions, emissions, and regimes together

$$p(\underbrace{A}_{\text{transitions}}, \underbrace{\phi}_{\text{emissions}}, \underbrace{s_{1:T}}_{\text{regimes}} \mid y) \propto p(y \mid s, \phi) p(s \mid A) \underbrace{p(A)}_{\text{Dirichlet}} p(\phi)$$

The pieces

- The transition matrix, with a Dirichlet prior on each row.
- The emission parameters for each regime.
- The hidden regime sequence itself.

How it's sampled

- The same Gibbs strategy as the drift models.
- Draw the regime path, then the parameters, and repeat.
- FFBS has a discrete cousin for the regime path.

From Baum-Welch to a Posterior

A point fit becomes a distribution

The old fit

- Baum-Welch maximizes the likelihood.
- It returns one transition matrix and one set of parameters.
- Uncertainty about them is invisible.

The Bayesian version

- Priors regularize small-sample regime estimates.
- The posterior spreads over plausible transition matrices.
- That spread flows into the regime probabilities.

Bayesian Regime Detection

Credible regime probabilities, not hard labels

The point of the Bayesian HMM is a softer, more honest answer to 'what regime are we in?'. Instead of a hard label that flips back and forth, you get a probability of crisis at each date — with a credible band around it.

The hard-label problem

- A threshold turns probabilities into a 0/1 label.
- Near the threshold, it flip-flops noisily.
- It looks decisive but is often just noise.

The soft alternative

- Report the probability of each regime directly.
- Add a credible band from the posterior.
- Decisiveness now reflects real evidence.

Soft Probability vs Hard Label

Two ways to answer the regime question

A hard label says

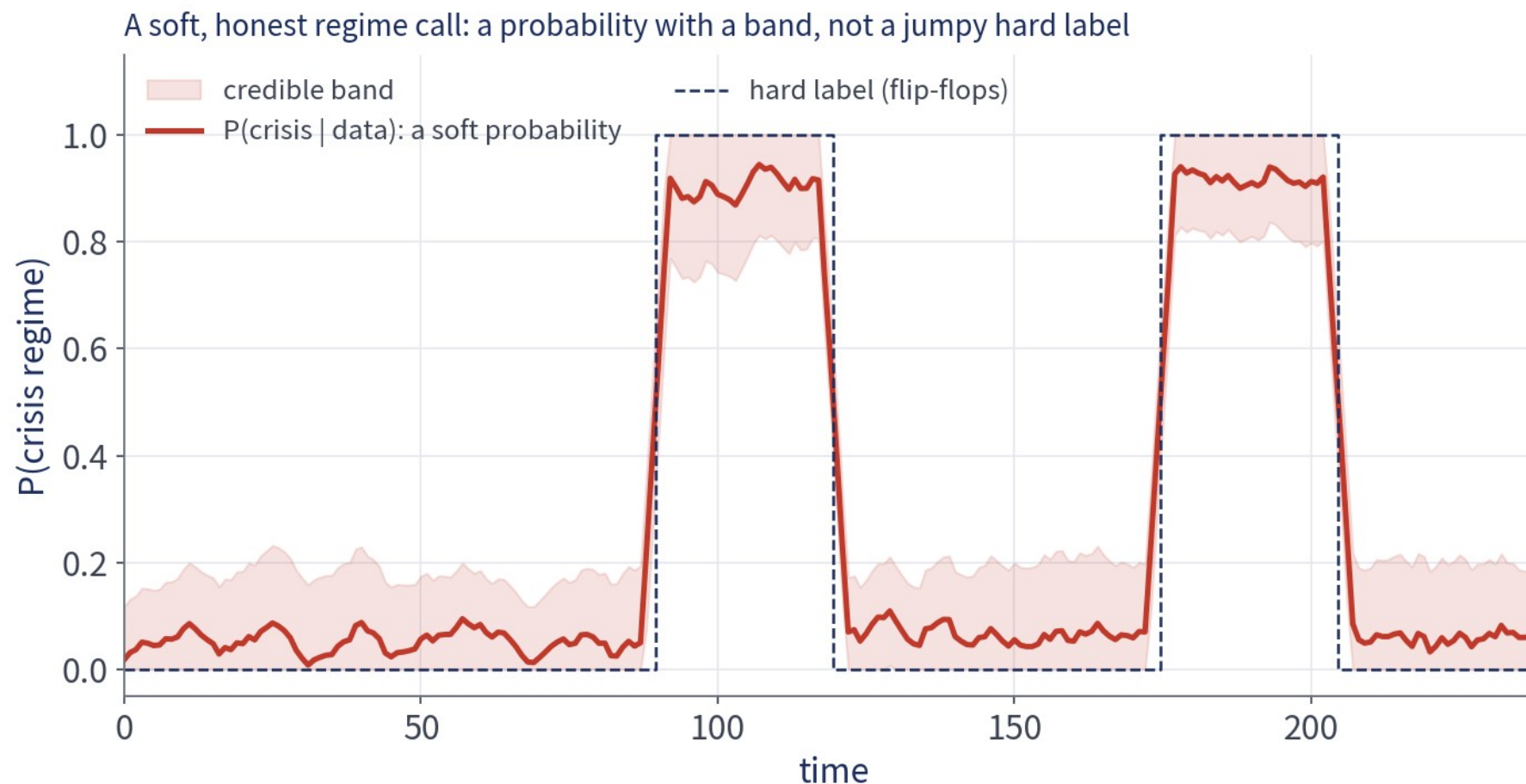
- 'We are in a crisis' — or not.
- One bit, no sense of confidence.
- It jumps whenever the estimate crosses a line.

A soft probability says

- 'There is a 70% chance we are in a crisis.'
- It carries its own confidence.
- It moves smoothly as evidence accumulates.

A Soft, Honest Regime Call

The probability of crisis, as a smooth posterior curve with a credible band, versus a hard 0/1 label that flip-flops. The soft call tells you not just which regime, but how sure the model is.



Why Soft Probabilities Beat Hard Labels

Especially at the edges

For understanding

- You see the ramp into and out of a regime.
- Ambiguous periods look ambiguous, not certain.
- You are never fooled by a confident-looking flip.

For decisions

- You can scale exposure by the probability.
- No arbitrary threshold to trip over.
- Uncertainty flows straight into position size.

Structural Break Points

When the parameters jump at an unknown date

Sometimes the world doesn't cycle between regimes — it changes once and stays changed. A structural break is a one-time jump in the parameters at an unknown date. The Bayesian approach puts a posterior on WHEN the break happened.

What a break is

- The model's parameters shift at a single date.
- Before and after look like different regimes.
- But it happens once, not repeatedly.

The Bayesian question

- Treat the break date as an unknown parameter.
- Infer a posterior over when it occurred.
- Get both whether AND when, with uncertainty.

The Break Date as an Unknown

Put a posterior on 'when'

The idea

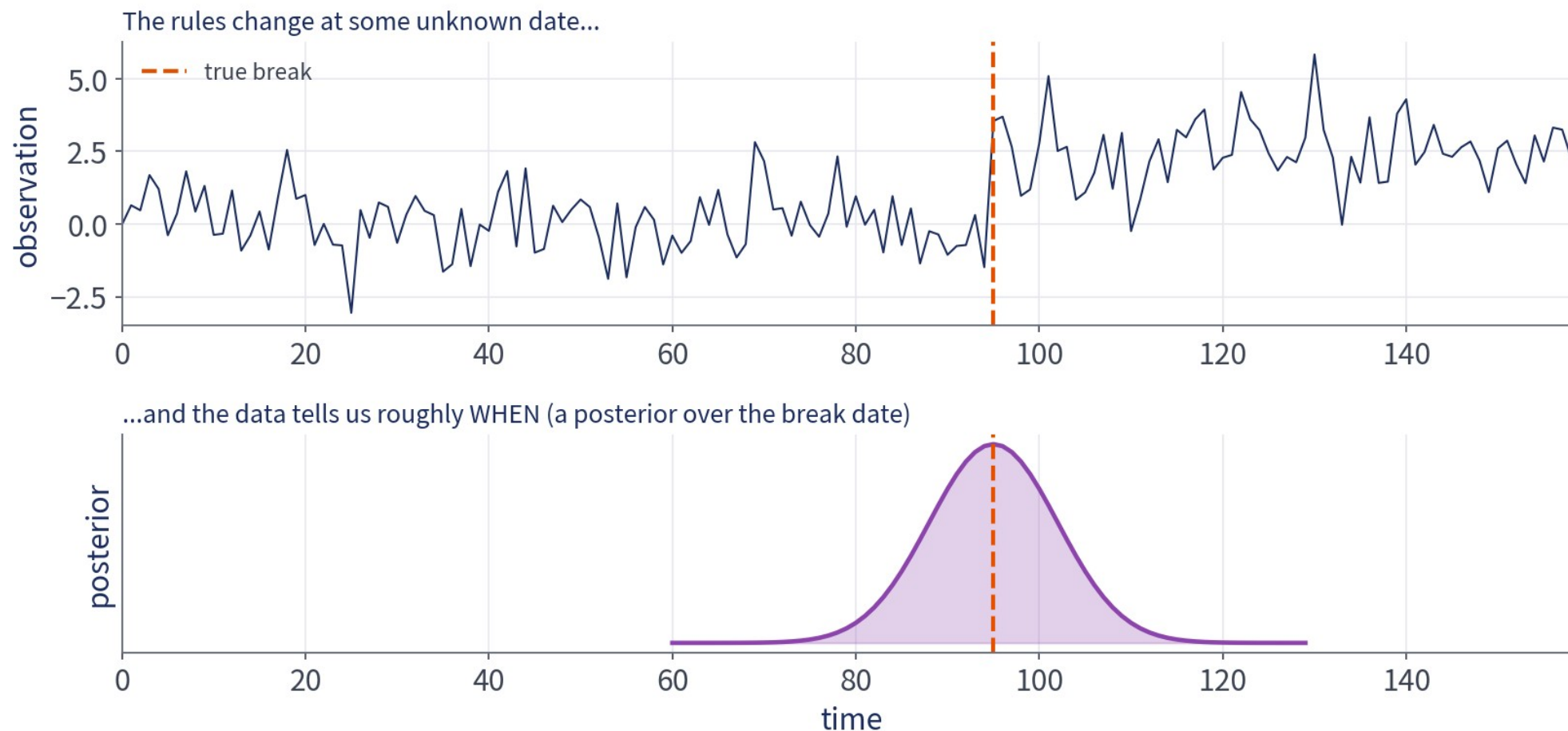
- The change-point is just another parameter.
- The data is more likely under the right split.
- So the posterior concentrates near the true date.

What you learn

- A most-probable break date — and a range around it.
- Whether the break is sharp or the date is fuzzy.
- Honest uncertainty about the timing itself.

A Structural Break With an Uncertain Date

The series shifts at some unknown time (top). The data tells us roughly when: a posterior over the break date (bottom) peaks near the true change but spreads to reflect our uncertainty about exactly when.



Regimes vs Breaks

Recurring states or a one-time jump

Regimes (HMM)

- The world cycles among a few recurring states.
- Calm and crisis come and go, and come back.
- Modeled by a hidden Markov chain.

Structural breaks

- A single, permanent shift in the rules.
- No return to the old regime.
- Modeled by a change-point at an unknown date.

A Bayesian HMM Gotcha: Label Switching

The regimes have no inherent names

A subtle trap: the model doesn't know which regime is 'regime 1'. Swapping the labels leaves the likelihood unchanged, so during sampling the chain can relabel the regimes. Average the raw draws blindly, and the regimes blur into each other.

Why it happens

- The likelihood is identical under any relabeling.
- So 'regime 1' and 'regime 2' can trade places mid-chain.
- The names are arbitrary; the model can't tell them apart.

The symptom

- Averaging a regime's parameter mixes both regimes.
- The two regimes collapse toward the same values.
- Your posterior looks meaningless.

Handling Label Switching

Pin the labels down

Simple remedies

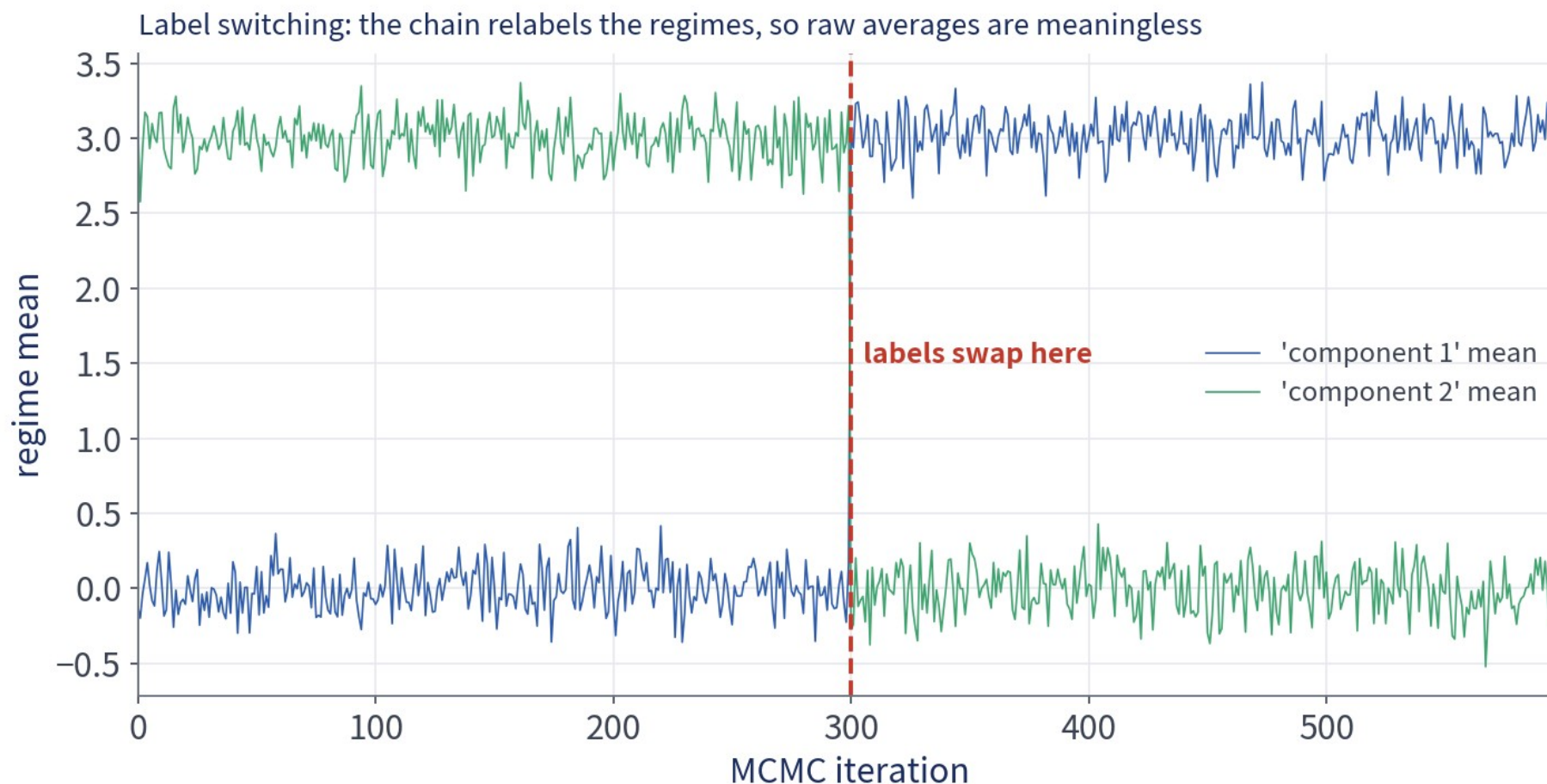
- Impose an ordering, e.g. regime means increasing.
- That breaks the symmetry and fixes the names.
- Often enough for two or three regimes.

More careful remedies

- Relabel the draws after sampling to align them.
- Then summarize each regime consistently.
- Always check the traces for swaps before trusting summaries.

Label Switching in Action

The trace of 'component 1's' mean jumps between two values partway through: the chain has relabeled the regimes. A naive average would land between them and describe neither regime correctly.



From Posteriors to Position Sizes

Letting uncertainty flow all the way through

All this honesty matters only if it changes what you do. The final step lets the posterior flow into the trade: a strong edge with a tight band and a confident regime gets full size; a shaky edge with a wide band or an ambiguous regime gets scaled down.

What feeds the size

- The strength of the signal (the edge).
- The conviction: how probable a favorable regime is.
- The tightness of the credible band.

The result

- Big when sure, small when doubtful.
- No hard on/off switch to whipsaw you.
- Risk scaled to real, stated uncertainty.

Sizing by Conviction

One expression for 'size to your certainty'

$$\text{position}_t \propto \underbrace{\text{edge}_t}_{\text{signal}} \times \underbrace{P(\text{favorable} \mid y)}_{\text{conviction}} \times \underbrace{\frac{1}{\text{band width}}}_{\text{less doubt, more size}}$$

Reading it

- Start from the raw edge — the signal.
- Scale by the probability the regime is favorable.
- Shrink when the credible band is wide.

The behavior

- Full size only when signal, regime, and band all agree.
- Automatic caution when any of them is shaky.
- Uncertainty is not a footnote — it sets the bet.

Pairs Trading Stage VI: Trading With Uncertainty

The thread's completion in the core course

This is where the pairs-trading thread arrives. Stage VI trades the pair using everything: a hedge ratio with a credible band, a soft regime probability, and conviction-based sizing. You trade by how sure you are — not by a hard label.

What Stage VI uses

- The Bayesian hedge ratio and its band (Part 2).
- The soft regime probability (this part).
- Conviction-based position sizing.

How it trades

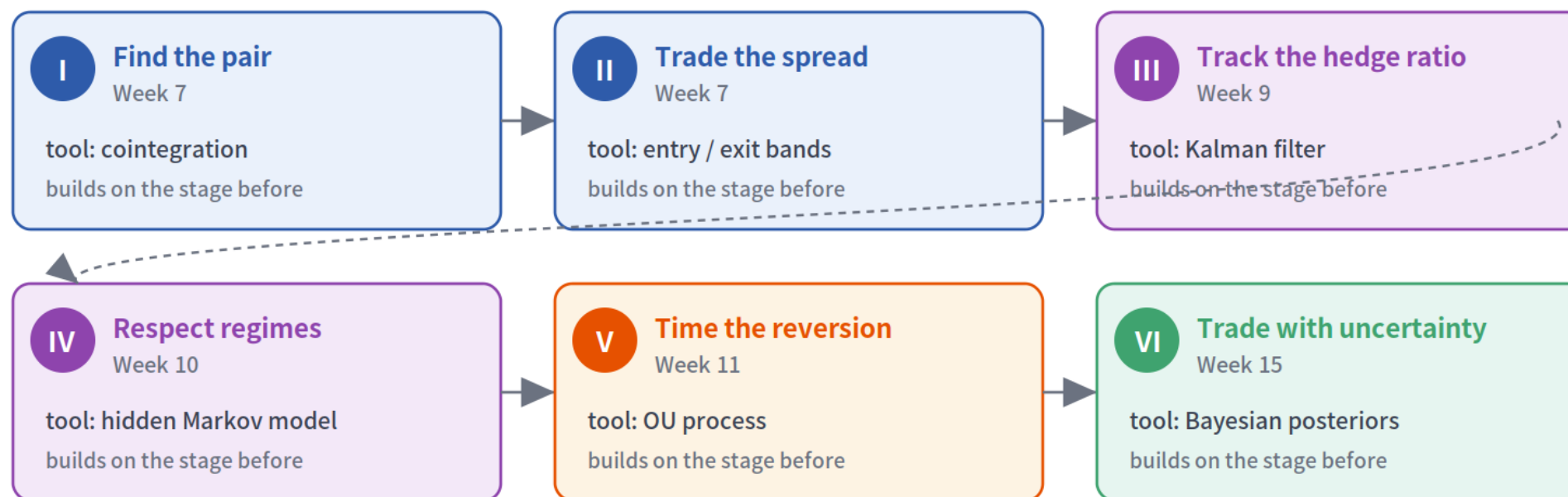
- Enter firmly when everything aligns and is tight.
- Trade small, or stand aside, when doubtful.
- Uncertainty is built into every decision.

The Pairs-Trading Thread, Assembled

Six stages, one evolving strategy: find the pair, trade the spread, track the drifting hedge ratio, respect regimes, time the mean reversion, and finally trade with honest uncertainty. Stage VI completes the core course.

The Pairs-Trading Thread, Assembled

Six stages, one evolving strategy - Stage VI completes the core course.



Stage VI adds the missing ingredient - honest uncertainty - and closes the core course.

Stage VI in Practice

Trade by conviction, not by a label

The old way

- A hard signal said in or out.
- One hedge ratio, taken as exact.
- The same size regardless of confidence.

The Stage VI way

- A probability of a good setup, not a yes/no.
- A hedge ratio with a band you can see.
- Size that rises and falls with conviction.

Bayesian Regimes for the Pair (Stage VI)

Direct Claude Code to trade by conviction

GOAL Fit a Bayesian regime model to the pair and size the trade by conviction, not a hard label.

[INPUT] Prompt · Claude Code

Write and run stage6_pair.py that:

1. fits a two-regime Bayesian HMM to a pair's spread (calm vs stressed), with priors;
2. imposes an ordering to avoid label switching;
3. outputs a soft $P(\text{favorable regime})$ with a band, not a hard 0/1 label;
4. sizes the position by edge times conviction, scaled down when the band is wide.

Compare cumulative results to a hard-label rule and show the conviction-sized version is steadier.

How to work with it

- Order the regimes to fix labels.
- Report probabilities, not labels.
- Size by conviction and band width.
- Smaller when the model is unsure.

Deliverable: stage6_pair.py + soft regime probabilities and conviction-sized results vs a hard rule.

Three Questions Before You Finish

If you can answer these, Week 15 has landed

Q1	What does a Bayesian HMM give instead of a hard regime label?	A soft probability with a band
Q2	What is label switching?	The chain relabels the regimes
Q3	What three things set the Stage VI position size?	Edge, conviction, band width

All three come from soft regime calls, the label-switching gotcha, and conviction-based sizing.

Three Pitfalls to Avoid

Where Bayesian switching goes wrong

Common mistakes

- "A hard regime label is the honest answer."
- "I can average the raw HMM draws directly."
- "A structural break happened on a known date."

The reality

- A soft probability with a band is more honest.
- Fix label switching first, or the regimes blur together.
- The break date is itself uncertain — infer it.

Week 15 in Three Ideas

Pulling the whole week together

1

Point estimates hide uncertainty; posteriors reveal it

Parts 1-2: a filtered line omits parameter uncertainty. A Bayesian state-space model, sampled with FFBS, turns it into an honest fan — a hedge ratio and even a volatility with credible bands.

2

Switching, done honestly

Part 3: the Bayesian HMM gives soft regime probabilities, not flip-flopping labels; break models locate jumps at uncertain dates. Mind label switching before summarizing.

3

Uncertainty flows to the trade

Conviction-based sizing scales exposure by edge, regime probability, and band width. Pairs Stage VI trades by how sure you are — completing the thread and the core course.

Next — Week 16: Synthesis and the Final Project, where the whole course comes together.